

Inter VLAN Routing:

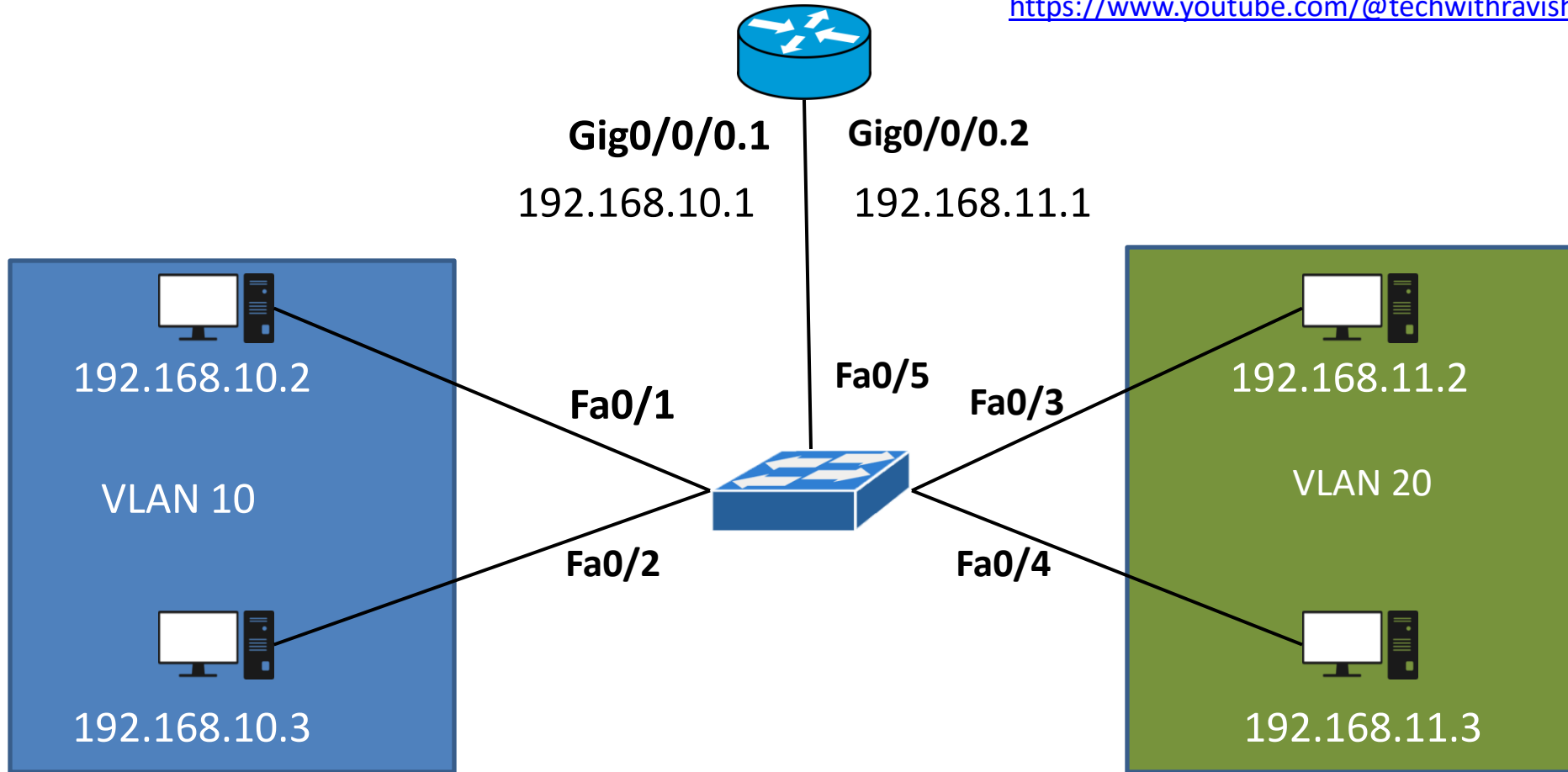
Inter-VLAN Routing refers to the process of routing traffic between different VLANs (Virtual Local Area Networks). VLANs are used to segment a larger network into smaller, isolated sub-networks, which enhances network performance and security. However, devices in different VLANs cannot communicate with each other without a Layer 3 device (such as a router or a Layer 3 switch) to route the traffic between them.

Why Inter-VLAN Routing is Needed

- Network Segmentation
- Security
- Communication

Router on a Stick: Inter VLAN Routing

<https://www.youtube.com/@techwithravish>



IVR Configuration:

Creating VLANs on Switch:

```
Switch> enable
```

```
Switch# configure terminal
```

```
Switch(config)# vlan 10
```

```
Switch(config-vlan)# name Sales
```

```
Switch(config-vlan)# exit
```

```
Switch(config)# vlan 20
```

```
Switch(config-vlan)# name Marketing
```

```
Switch(config-vlan)# exit
```

Configuring VLAN on Switch Interfaces

```
Switch> enable
```

```
Switch# configure terminal
```

```
Switch(config)# interface range fa0/1-2
```

```
Switch(config-if)#Switchport Mode Access
```

```
Switch(config-if)#Switchport Access VLAN 10
```

```
Switch(config-if)#Exit
```

```
Switch(config)# interface range fa0/3-4
```

```
Switch(config-if)#Switchport Mode Access
```

```
Switch(config-if)#Switchport Access VLAN 20
```

```
Switch(config-if)#)# exit
```

IVR Configuration: Router on a Stick

Configuring Trunk on Switch:

```
Switch>Enable
```

```
Switch#Configure terminal
```

```
Switch(config)#Interface gigabitEthernet 0/1
```

```
Switch(config-if)#Switchport mode trunk
```

```
Switch(config-if)#Switchport trunk allowed vlan 10,20
```

```
Switch(config-if)#Exit
```

IVR Configuration:

```
Router>Enable
```

```
Router#Configure terminal
```

```
Router(config)#Interface gigabitEthernet 0/0/0.1
```

```
Router(config-subif)#Encapsulation dot1Q 10
```

```
Router(config-subif)#IP address 192.168.10.1 255.255.255.0
```

```
Router(config-subif)#Exit
```

```
Router(config)#Interface GigabitEthernet 0/0/0.2
```

```
Router(config-subif)#Encapsulation dot1Q 20
```

```
Router(config-subif)#Ip address 192.168.20.1 255.255.255.0
```

```
Router(config-subif)#Exit
```